ASSIGNMENT 4-5 REPORT

Incident Response Simulation and
Infrastructure as Code Implementation

MicroShop - Containerized Microservices System

Introduction to SRE | 2026

GitHub: https://github.com/Nurlybek-Nurzhan/Assignment_4-5_SRE

System Overview

MicroShop is a containerized e-commerce platform built to demonstrate SRE practices. It consists of five independent FastAPI microservices, a PostgreSQL database, an Nginx frontend acting as a reverse proxy, Prometheus for metrics collection with three alert rules, and Grafana with an auto-provisioned dashboard. All components run in Docker containers orchestrated by Docker Compose on a shared internal bridge network (microshop-net).

Architecture

- Auth Service (port 8001) - JWT authentication with bcrypt password hashing; crashes on startup if JWT_SECRET not set
- Product Service (port 8002) - product catalog, persisted to a Docker volume via JSON file
- Order Service (port 8003) - order management, PostgreSQL backend, Histogram latency metrics; crashes if DB_PASS or JWT_SECRET not set
- User Service (port 8004) - user profile management with JWT token verification; crashes if JWT_SECRET not set
- Chat Service (port 8005) - messaging with JWT token verification; crashes if JWT_SECRET not set
- PostgreSQL - persistent database used exclusively by the Order Service
- Prometheus (port 9090) - scrapes /metrics from all five services; three alert rules defined
- Grafana (port 3000) - auto-provisioned datasource and dashboard on startup
- Nginx (port 80) - serves the frontend, proxies /api/* routes to backend services

Observability and Metrics

All five services expose Prometheus Counter metrics for request counts. The Order Service additionally exposes a Histogram metric - order_request_duration_seconds - with nine latency buckets (5ms to 2.5s). This enables p50, p95, and p99 latency queries in Grafana, which are the primary signals used in production SLO monitoring.

Three Prometheus alert rules are defined in monitoring/alert_rules.yml:

- OrderServiceErrors - fires when order_errors_total increases (DB connection broken)
- ServiceDown - fires when any scrape target returns up=0 (process crashed)
- OrderServiceHighLatency - fires when p95 latency exceeds 1 second for 1 minute

Security Hardening

- All credentials stored in .env file, never hardcoded in source
- Passwords hashed with bcrypt - plaintext never stored
- Real JWT tokens (HS256) issued on login; Chat, User services validate on every request
- Services crash at startup with RuntimeError if JWT_SECRET or DB_PASS env vars are missing
- All containers have enforced memory and CPU limits (mem_limit / cpus top-level keys)
- Docker healthchecks on all services - Docker knows if app is ready, not just started
- terraform.tfstate excluded from git - contains sensitive AWS resource IDs
- allowed_ssh_cidr has no default in variables.tf - terraform apply fails if not explicitly set

Known Limitations (Assignment Scope)

- Auth service uses two hardcoded users defined in .env - no user registration flow
- Chat messages are in-memory only - reset on container restart (no chat DB)
- User list is a static in-memory list - no persistence across restarts
- Order Service opens a new psycopg2 connection per request - no connection pooling
- allowed_ssh_cidr is set to 0.0.0.0/0 in terraform.tfvars for demo purposes - must be restricted to a specific IP in production

Running System

```
PROBLEMS   DEBUG CONSOLE   OUTPUT   TERMINAL   PORTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
time="2026-04-29T16:10:58+05:00" level=warning msg="C:\\Users\\Nurzhan\\Favorites\\Basa\\Education\\Programming\\Introduction to SRE\\assignment-4-5\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, pl
ease remove it to avoid potential confusion"
[+] up 10/10
 ✓ Network assignment-4-5_microshop-net Created                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.1s
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 ✓ Container prometheus                 Created                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.8s
 ✓ Container grafana                    Created                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.8s
 ✓ Container user-service               Created                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.7s
 ✓ Container frontend                   Created                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.8s
 ✓ Container product-service            Created                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.6s
 ✓ Container auth-service               Created                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.6s
 ✓ Container chat-service               Created                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.7s
 ✓ Container order-service              Created                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
Attaching to auth-service, chat-service, frontend, grafana, order-service, postgres, product-service, prometheus, user-service
postgres  |
postgres  | PostgreSQL Database directory appears to contain a database; Skipping initialization
postgres  |
postgres  | 2026-04-29 11:11:01.466 UTC [1] LOG:  starting PostgreSQL 15.17 on x86_64-pc-linux-musl, compiled by gcc (Alpine 15.2.0) 15.2.0, 64-bit
postgres  | 2026-04-29 11:11:01.466 UTC [1] LOG:  listening on IPv4 address "0.0.0.0", port 5432
postgres  | 2026-04-29 11:11:01.466 UTC [1] LOG:  listening on IPv6 address "::", port 5432
postgres  | 2026-04-29 11:11:01.496 UTC [1] LOG:  listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
postgres  | 2026-04-29 11:11:01.540 UTC [29] LOG:  database system was shut down at 2026-04-29 11:10:44 UTC
postgres  | 2026-04-29 11:11:01.569 UTC [1] LOG:  database system is ready to accept connections
user-service  | INFO:     Started server process [1]
user-service  | INFO:     Waiting for application startup.
user-service  | INFO:     Application startup complete.
user-service  | INFO:     Uvicorn running on http://0.0.0.0:8004 (Press CTRL+C to quit)
product-service  | INFO:     Started server process [1]
product-service  | INFO:     Waiting for application startup.
product-service  | INFO:     Application startup complete.
product-service  | INFO:     Uvicorn running on http://0.0.0.0:8002 (Press CTRL+C to quit)
auth-service     | INFO:     Started server process [1]
auth-service     | INFO:     Waiting for application startup.


auth-service     | INFO:     Uvicorn running on http://0.0.0.0:8001 (Press CTRL+C to quit)

chat-service     | INFO:     Application startup complete.
chat-service     | INFO:     Uvicorn running on http://0.0.0.0:8005 (Press CTRL+C to quit)
Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint grafana (b4a0f305c16adaa710896e265da9eb0c4e53a3b6a9f0392acab75071d368c021): failed to bind host port 0
.0.0.0:3000/tcp: address already in use
 v View in Docker Desktop   o View Config   w Enable Watch   d Detach
Nurzhan@Nurzhan MINGW64 ~/Favorites/Basa/Education/Programming/Introduction to SRE/assignment-4-5
$ ⊓
```

Figure: docker compose up - all 9 containers started successfully

```
docker compose up -d
time="2026-04-29T16:42:16+05:00" level=warning msg="C:\\Users\\Nurzhan\\Favorites\\Basa\\Education\\Programming\\Introduction to SRE\\assignment-4-5\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, pl
ease remove it to avoid potential confusion"
[+] up 9/9
 ✓ Container chat-service    Started                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                2.1ss
 ✓ Container postgres        Healthy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                12.5s
 ✓ Container user-service    Started                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                2.2ss
 ✓ Container grafana         Started                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                2.2ss
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 ✓ Container frontend        Started                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                1.9ss
 ✓ Container product-service Started                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                2.0ss
 ✓ Container auth-service    Started                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                1.9ss
 ✓ Container order-service   Started                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                12.4s

Nurzhan@Nurzhan MINGW64 ~/Favorites/Basa/Education/Programming/Introduction to SRE/assignment-4-5
$ docker ps
CONTAINER ID   IMAGE                      COMMAND                  CREATED          STATUS                    PORTS                                          NAMES
063f93add541   assignment-4-5-order-service   "uvicorn main:app --…"   24 seconds ago   Up 11 seconds             0.0.0.0:8003->8003/tcp, [::]:8003->8003/tcp    order-service
2a823a574beb   prom/prometheus:latest     "/bin/prometheus --c…"   25 seconds ago   Up 22 seconds             0.0.0.0:9090->9090/tcp, [::]:9090->9090/tcp    prometheus
249a23b67477   assignment-4-5-user-service    "uvicorn main:app --…"   25 seconds ago   Up 22 seconds             0.0.0.0:8004->8004/tcp, [::]:8004->8004/tcp    user-service
91a185d11b6a   nginx:alpine               "/docker-entrypoint.…"   25 seconds ago   Up 22 seconds             0.0.0.0:80->80/tcp, [::]:80->80/tcp            frontend
3674f2d5f94f   postgres:15-alpine         "docker-entrypoint.s…"   25 seconds ago   Up 22 seconds (healthy)   5432/tcp                                       postgres
38012b3b6dec   assignment-4-5-auth-service    "uvicorn main:app --…"   25 seconds ago   Up 22 seconds             0.0.0.0:8001->8001/tcp, [::]:8001->8001/tcp    auth-service
22da15968f20   assignment-4-5-product-service "uvicorn main:app --…"   25 seconds ago   Up 22 seconds             0.0.0.0:8002->8002/tcp, [::]:8002->8002/tcp    product-service
5751056d5922   grafana/grafana:latest     "/run.sh"                25 seconds ago   Up 22 seconds             0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp    grafana
0be8f4eda280   assignment-4-5-chat-service    "uvicorn main:app --…"   25 seconds ago   Up 22 seconds             0.0.0.0:8005->8005/tcp, [::]:8005->8005/tcp    chat-service
```

Figure: docker ps - all containers running with correct port mappings

Frontend Dashboard

The MicroShop Dashboard shows real-time health of all five services with green/red status dots. The header includes a login form - entering credentials calls the Auth Service and receives a real JWT token which is then used for authenticated Chat and User Service calls.

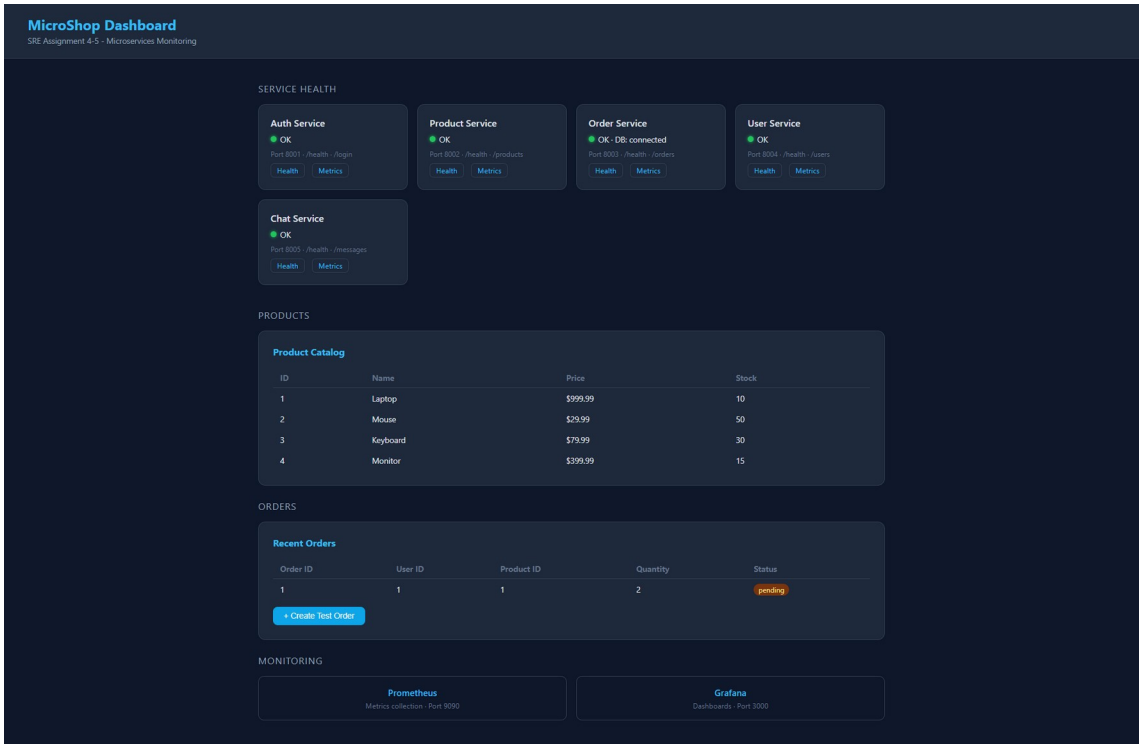

Figure: MicroShop Dashboard - all 5 services healthy, login form in header

Monitoring

Prometheus scrapes all five services every 15 seconds. Grafana loads the Prometheus datasource and MicroShop dashboard automatically on startup via provisioning files - no manual configuration needed. The dashboard includes panels for service health, per-service request rates, order error rate, total error count, and order latency percentiles.

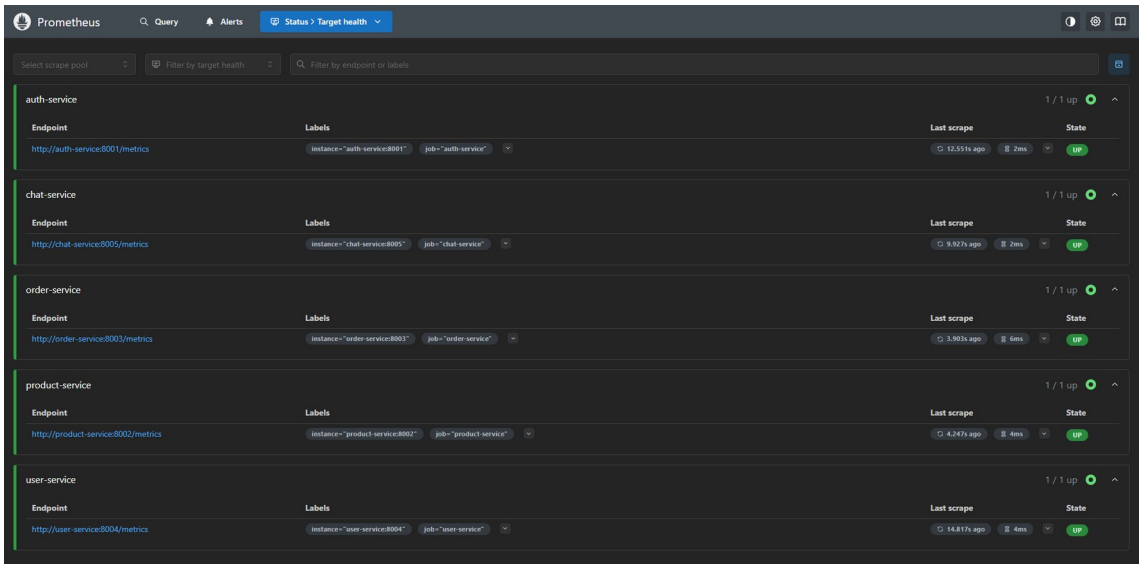

Figure: Prometheus - all 5 service targets showing UP

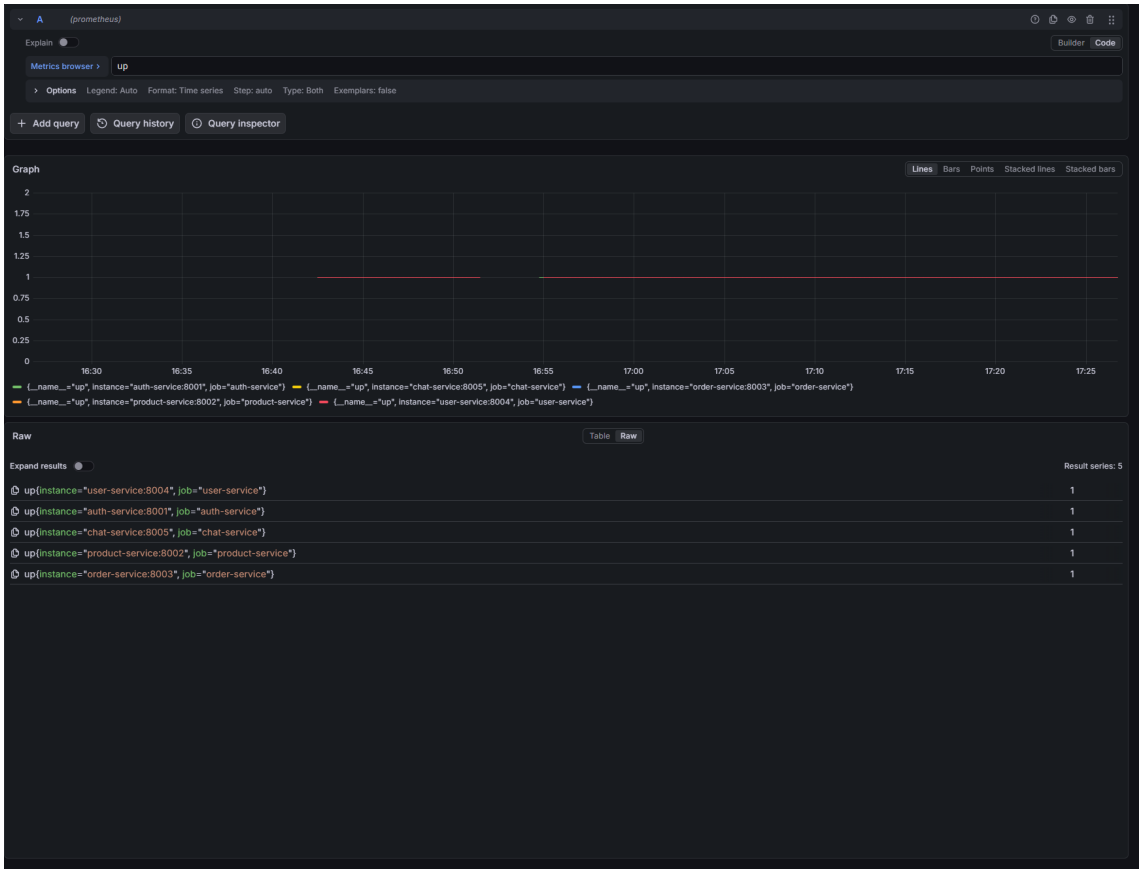

Figure: Grafana - auto-provisioned dashboard with health, request rate, and error panels

ASSIGNMENT 4 - Incident Report

1. Incident Summary

On 2026-04-29, a production-level incident was simulated in the MicroShop Order Service by deliberately misconfiguring the DB_HOST environment variable from 'postgres' to 'wrong-host'. After restarting the container with this broken configuration, the Order Service lost all database connectivity and returned HTTP 503 on every request. All four other services remained fully operational throughout the incident, demonstrating correct fault isolation.

2. Impact Assessment

- Order Service: 100% unavailable - /orders (GET and POST) and /health all returned HTTP 503
- Frontend dashboard: Order Service card showed 'Unreachable' with red status dot
- Users could not view existing orders or create new ones
- Auth, Product, User, and Chat services: fully operational - fault isolation confirmed
- Data integrity: no data lost - PostgreSQL remained healthy throughout
- Prometheus showed order-service as up=1 - the process was running but functionally broken

3. Severity Classification

Severity: SEV-2 (Major) - a critical business function (order processing) was completely unavailable.

Classified SEV-2 rather than SEV-1 because the failure was isolated to one service. Authentication, product browsing, and other features remained functional.

4. Timeline of Events

Time	Event
T+00:00	DB_HOST changed from 'postgres' to 'wrong-host' in docker-compose.yml
T+00:05	Order Service restarted: docker compose up -d --force-recreate order-service
T+00:10	Frontend dashboard detected Order Service as 'Unreachable'
T+00:15	/api/orders/orders returned HTTP 503 Service Unavailable
T+00:20	order_errors_total counter began incrementing in Prometheus
T+00:25	/health returned 503: 'DB connection failed: could not translate host name wrong-host'
T+00:30	docker logs confirmed repeated psycopg2 DNS resolution failures
T+01:00	Root cause confirmed: incorrect DB_HOST environment variable
T+01:05	DB_HOST corrected back to 'postgres' in docker-compose.yml
T+01:10	Order Service restarted with corrected configuration
T+01:15	/health returned HTTP 200: {status: ok, db: connected}
T+01:20	Frontend dashboard confirmed all services green - full recovery

5. Root Cause Analysis

Root cause: misconfigured DB_HOST environment variable. Docker Compose services resolve each other by container name through Docker's internal DNS. The hostname 'postgres' resolves correctly on the microshop-net network. The value 'wrong-host' has no DNS record, so every call to psycopg2.connect() immediately raised an OperationalError at the DNS resolution step before attempting any TCP connection.

Key SRE observation: Prometheus showed order-service as up=1 throughout the incident. Prometheus scrapes /metrics, which does not touch the database and always responds. The service process was alive but functionally broken. This is the critical SRE distinction: process liveness does not equal service health. The /health endpoint correctly returned 503 with the exact error message. The order_request_duration_seconds Histogram also showed latency spikes before failures, which would have triggered the OrderServiceHighLatency alert in a real environment.

Incident Evidence

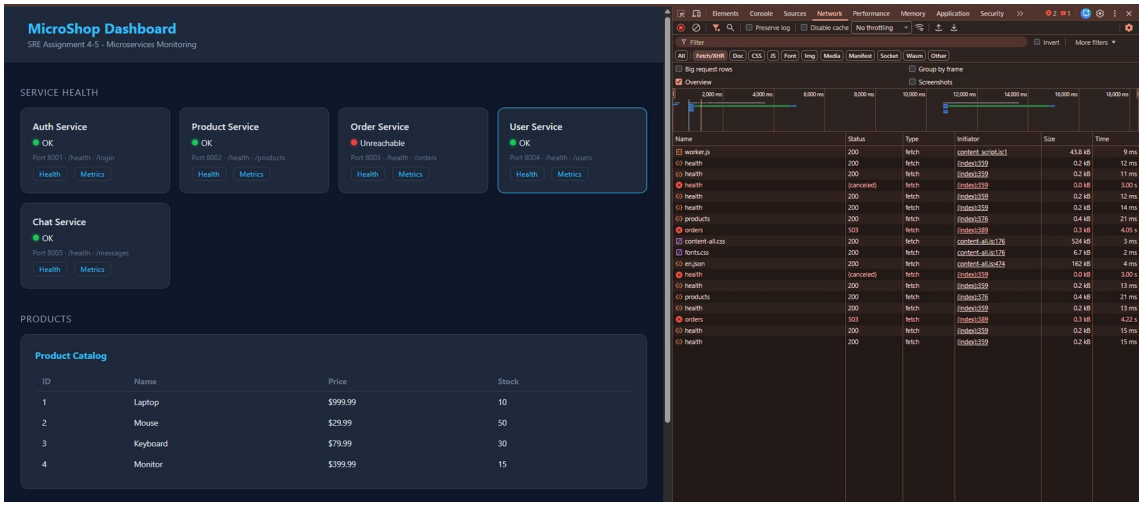

Figure: Dashboard during incident - Order Service 'Unreachable', 503 errors in Network tab

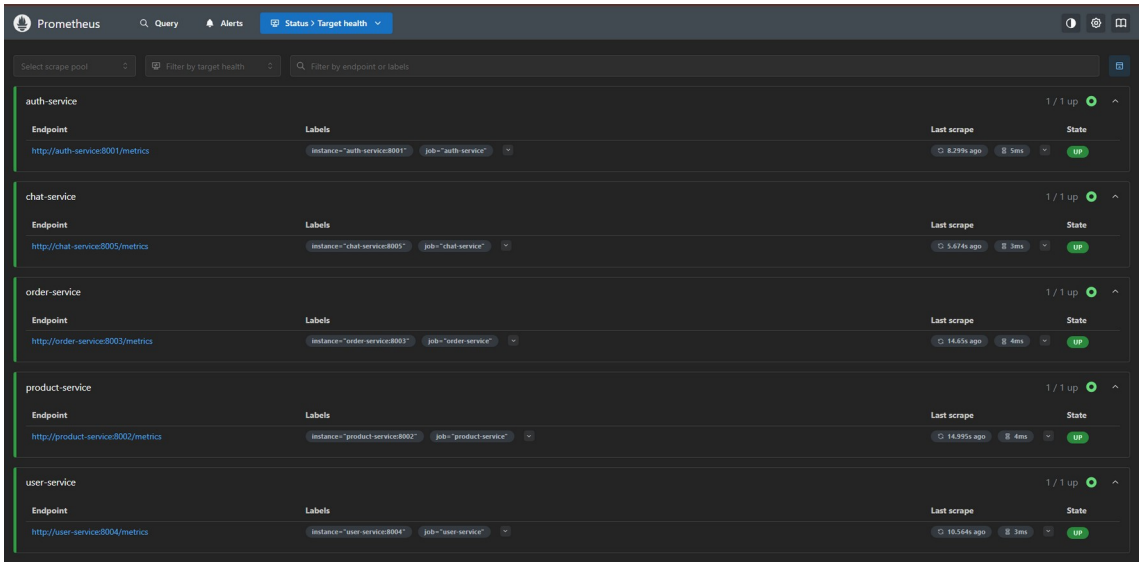

Figure: Prometheus during incident - up=1 for all services. Process alive but DB broken. Real signal: order_errors_total incrementing.

```
Nurzhan@Nurzhan MINGW64 ~/Favorites/Basa/Education/Programming/Introduction to SRE/assignment-4-5
$ docker logs order-service 2>&1 | tail -20
INFO:     172.19.0.3:52208 - "GET /orders HTTP/1.1" 503 Service Unavailable
INFO:     172.19.0.9:57344 - "GET /metrics HTTP/1.1" 200 OK
INFO:     172.19.0.3:52114 - "GET /orders HTTP/1.1" 503 Service Unavailable
INFO:     172.19.0.9:42486 - "GET /metrics HTTP/1.1" 200 OK
INFO:     172.19.0.3:43896 - "GET /orders HTTP/1.1" 503 Service Unavailable
INFO:     172.19.0.3:40350 - "GET /orders HTTP/1.1" 503 Service Unavailable
INFO:     172.19.0.9:34554 - "GET /metrics HTTP/1.1" 200 OK
INFO:     172.19.0.3:37070 - "GET /orders HTTP/1.1" 503 Service Unavailable
INFO:     172.19.0.9:42780 - "GET /metrics HTTP/1.1" 200 OK
INFO:     172.19.0.3:55036 - "GET /orders HTTP/1.1" 503 Service Unavailable
INFO:     172.19.0.3:51186 - "GET /orders HTTP/1.1" 503 Service Unavailable
INFO:     172.19.0.9:49238 - "GET /metrics HTTP/1.1" 200 OK
INFO:     172.19.0.3:54910 - "GET /orders HTTP/1.1" 503 Service Unavailable
INFO:     172.19.0.9:33888 - "GET /metrics HTTP/1.1" 200 OK
INFO:     172.19.0.3:58512 - "GET /orders HTTP/1.1" 503 Service Unavailable
INFO:     172.19.0.3:53038 - "GET /orders HTTP/1.1" 503 Service Unavailable
INFO:     172.19.0.9:34342 - "GET /metrics HTTP/1.1" 200 OK
INFO:     172.19.0.3:50634 - "GET /orders HTTP/1.1" 503 Service Unavailable
INFO:     172.19.0.9:55712 - "GET /metrics HTTP/1.1" 200 OK
INFO:     172.19.0.3:58176 - "GET /orders HTTP/1.1" 503 Service Unavailable

Nurzhan@Nurzhan MINGW64 ~/Favorites/Basa/Education/Programming/Introduction to SRE/assignment-4-5
$
```

Figure: Docker logs - GET /orders returns 503, GET /metrics returns 200. Confirms service running but every DB request fails.

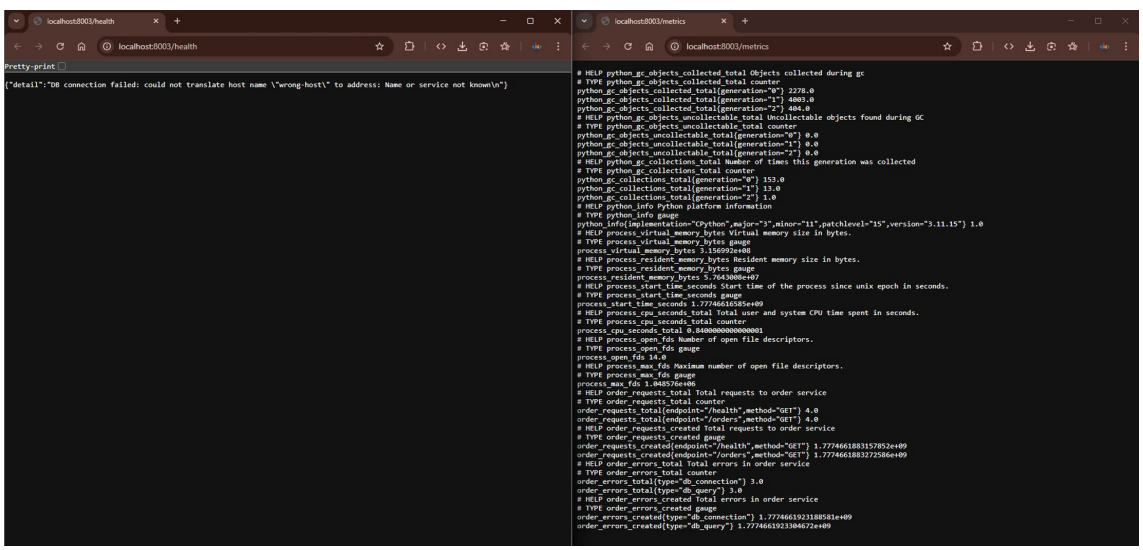

Figure: /health returning 503 with exact error: 'could not translate host name wrong-host' - root cause evidence.

6. Mitigation Steps

- Step 1 - docker logs order-service showed repeated 503 on /orders
- Step 2 - /health endpoint returned exact error with the bad hostname 'wrong-host'
- Step 3 - Inspected docker-compose.yml: DB_HOST was set to 'wrong-host'
- Step 4 - Changed DB_HOST back to 'postgres'
- Step 5 - Restarted only the affected container: docker compose up -d --force-recreate order-service
- Step 6 - Verified: /health returned 200 with db: connected
- Step 7 - Confirmed via dashboard: all services green

7. Resolution Confirmation

Full recovery achieved within approximately 70 seconds of applying the fix. /health returned HTTP 200, the dashboard showed all five services green, and the order_errors_total counter stopped incrementing. No data was lost.

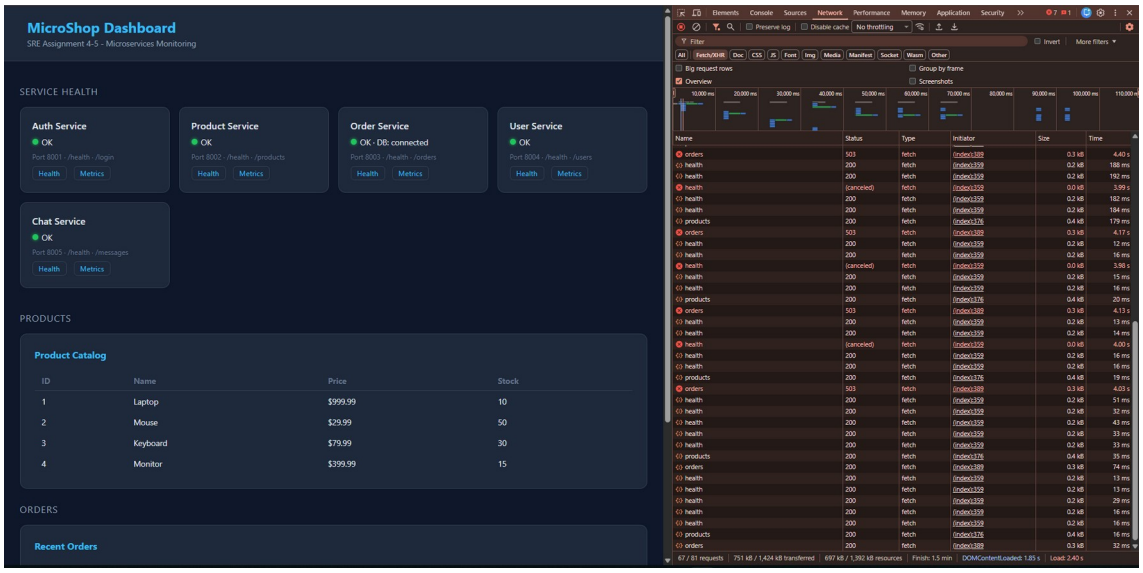

Figure: Recovery - all services green, Order Service showing DB: connected

```
Nurzhan@Nurzhan MINGW64 ~/Favorites/Basa/Education/Programming/Introduction to SRE/assignment-4-5
$ docker logs order-service 2>&1 | tail -10
INFO:     172.19.0.3:41564 - "GET /orders HTTP/1.1" 200 OK
INFO:     172.19.0.9:46448 - "GET /metrics HTTP/1.1" 200 OK
INFO:     172.19.0.3:36050 - "GET /health HTTP/1.1" 200 OK
INFO:     172.19.0.3:36060 - "GET /orders HTTP/1.1" 200 OK
INFO:     172.19.0.9:56206 - "GET /metrics HTTP/1.1" 200 OK
INFO:     172.19.0.3:46088 - "GET /health HTTP/1.1" 200 OK
INFO:     172.19.0.3:46094 - "GET /orders HTTP/1.1" 200 OK
INFO:     172.19.0.3:54568 - "GET /health HTTP/1.1" 200 OK
INFO:     172.19.0.3:54578 - "GET /orders HTTP/1.1" 200 OK
INFO:     172.19.0.9:51814 - "GET /metrics HTTP/1.1" 200 OK

Nurzhan@Nurzhan MINGW64 ~/Favorites/Basa/Education/Programming/Introduction to SRE/assignment-4-5
$
```

Figure: Recovery logs - HTTP 200 responses confirmed after fix

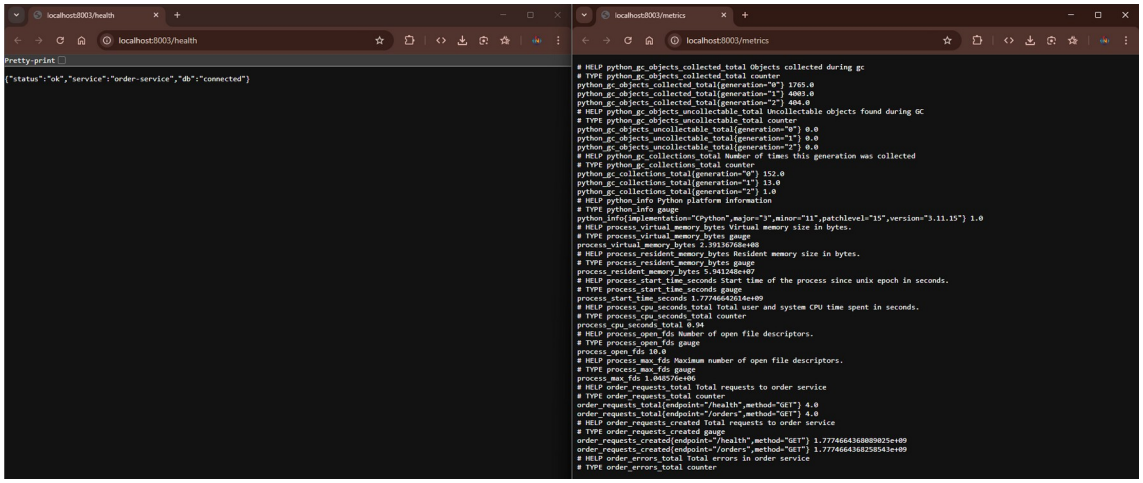

Figure: /health returning 200 with db: connected

Postmortem Analysis

1. Incident Overview

On 2026-04-29, the MicroShop Order Service became fully unavailable for approximately 70 seconds due to an incorrect database hostname in the service environment configuration. The incident was intentionally introduced as part of Assignment 4 to simulate a realistic production failure and practice the complete incident response lifecycle.

2. Customer Impact

- 100% of order-related requests failed with HTTP 503 during the incident window
- Order viewing and creation were completely unavailable
- Product browsing, authentication, user profiles, and chat were fully unaffected
- No data loss - PostgreSQL remained healthy and all existing orders were preserved

3. Root Cause Analysis

The DB_HOST environment variable was manually changed to a non-existent hostname. Docker Compose services discover each other via Docker's internal DNS using container names. Setting DB_HOST='wrong-host' caused immediate DNS failure on every database operation. The service had no retry logic - each request attempted a fresh connection, failed instantly, incremented order_errors_total, and returned HTTP 503.

4. Detection and Response Evaluation

Detection was fast - the frontend dashboard showed the failure within 10 seconds. The /health endpoint was the critical diagnostic tool: it returned the exact bad hostname in the error message, making root cause identification immediate.

Gap identified during incident: Prometheus showed up=1 throughout because it only checks process liveness via /metrics. The OrderServiceErrors alert rule fires on error counter increases, and the new OrderServiceHighLatency alert fires on p95 latency exceeding 1 second - both would catch functional failures that process liveness misses.

5. Resolution Summary

A single configuration change (correcting DB_HOST to 'postgres') followed by a container restart fully restored the service in under 2 minutes. No code changes, no data recovery, no other service required intervention.

6. Lessons Learned

- Environment variables are a frequent source of misconfiguration in containerized systems
- Process liveness (up=1) does not mean a service is functionally healthy - dependency checks in /health are essential
- Histogram metrics (p95/p99 latency) provide earlier warning signals than counters alone
- Microservices fault isolation worked correctly - one broken service did not cascade to others
- Services now crash at startup if critical env vars are missing - fail fast beats silent wrong behavior

7. Action Items

- DONE - OrderServiceErrors alert: fires when order_errors_total increases
- DONE - ServiceDown alert: fires when any Prometheus target goes down
- DONE - OrderServiceHighLatency alert: fires when p95 latency exceeds 1 second
- DONE - Histogram metrics added to Order Service (order_request_duration_seconds)
- DONE - Services crash at startup if JWT_SECRET or DB_PASS not set
- DONE - allowed_ssh_cidr has no default - terraform apply requires explicit value
- TODO - Add DB connection retry with exponential backoff
- TODO - Configure Grafana notification channel (email/Slack) for alert delivery
- TODO - Restrict allowed_ssh_cidr to specific IP in production deployment

ASSIGNMENT 5 - Infrastructure as Code (Terraform + AWS)

Overview

The cloud infrastructure for MicroShop was provisioned using Terraform with the AWS provider. Terraform is a declarative IaC tool - the entire infrastructure is described in .tf files and created, updated, or destroyed with single commands. This makes infrastructure reproducible, version-controlled, and auditable.

Implementation: Two Phases

During development, Terraform was first configured with a local Docker provider (kreuzwerker/docker) to validate the configuration structure locally. This phase is visible in screenshots 13_terraform_apply_*. After the professor confirmed that cloud deployment was required, the configuration was migrated to the AWS provider. The final AWS deployment is shown in screenshots 14_terraform_apply_after_AWS_implementation_*. The code in the repository reflects the final AWS configuration only.

> Note: AWS apply created: EC2 instance at 34.238.137.187 (us-east-1), security group sg-0091acf407323802f.

Terraform Files

main.tf

Defines two AWS resources. aws_security_group opens inbound ports for SSH (22), HTTP (80), Grafana (3000), and Prometheus (9090). SSH and monitoring ports use the allowed_ssh_cidr variable - this variable has no default, so terraform apply fails if not explicitly set. aws_instance provisions a t3.micro EC2 VM with Amazon Linux 2. The user_data script installs Docker, git, and docker-compose on first boot. key_name accepts an optional SSH key pair via the ssh_key_name variable.

variables.tf

Declares five variables: aws_region, instance_type, ami_id, allowed_ssh_cidr (no default - forces explicit security decision), and ssh_key_name (optional, empty = no SSH key). Removing the default from allowed_ssh_cidr was a deliberate security hardening step - it prevents accidental open-internet deployments.

terraform.tfvars

Provides actual values: us-east-1, t3.micro, Amazon Linux 2 AMI, and allowed_ssh_cidr = 0.0.0.0/0 for this assignment demo. t3.micro was chosen because t2.micro was rejected by AWS as not free-tier eligible. The 0.0.0.0/0 value is intentional for demo access and documented with a comment explaining it must be restricted in production.

outputs.tf

Outputs after apply: instance_public_ip, instance_public_dns, grafana_url, prometheus_url, and security_group_id - making deployed services accessible without manually looking up the IP in the AWS Console.

Terraform Screenshots

```
Nurzhan@Nurzhan MINGW64 ~/Favorites/Basa/Education/Programming/Introduction to SRE/assignment-4-5/terraform
$ cd ~/Favorites/Basa/Education/Programming/Introduction\ to\ SRE/assignment-4-5/terraform
terraform init
Initializing the backend...
Initializing provider plugins...
- Reusing previous version of kreuzwerker/docker from the dependency lock file
- Finding hashicorp/aws versions matching "~> 5.0"...
- Using previously-installed kreuzwerker/docker v3.9.0
- Installing hashicorp/aws v5.100.0...
- Installed hashicorp/aws v5.100.0 (signed by HashiCorp)
Terraform has made some changes to the provider dependency selections recorded
in the .terraform.lock.hcl file. Review those changes and commit them to your
version control system if they represent changes you intended to make.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.

Nurzhan@Nurzhan MINGW64 ~/Favorites/Basa/Education/Programming/Introduction to SRE/assignment-4-5/terraform
$ []
```

Figure: terraform init - AWS provider downloaded successfully

```
Nurzhan@Nurzhan MINGW64 ~/Favorites/Basa/Education/Programming/Introduction to SRE/assignment-4-5/terraform
$ mkdir -p ~/.aws
cat > ~/.aws/credentials << 'EOF'
[default]
aws_access_key_id = AKIAUCXIRKVEJNVBJZ4V
aws_secret_access_key = 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
EOF

cat > ~/.aws/config << 'EOF'
[default]
region = us-east-1
EOF
```

Figure: AWS IAM credentials configured for Terraform authentication

```
PROBLEMS    DEBUG CONSOLE    OUTPUT    TERMINAL    PORTS

Nurzhan@Nurzhan MINGW64 ~/Favorites/Basa/Education/Programming/Introduction to SRE/assignment-4-5/terraform
$ terraform plan
docker_image.prometheus: Refreshing state... [id=sha256:dda13e28bf95a5e5ca5b8ed56852006094c1c8e8871d9c9dbeed30aa6e55271fprom/prometheus:latest]
docker_image.grafana: Refreshing state... [id=sha256:83749231c3835e390a3144e5e940203e42b9589761f20ef3169c716e734ad505grafana/grafana:latest]
docker_image.postgres: Refreshing state... [id=sha256:09e4f20b14ddb3dfe3a0c825b206032aaf8f28300ba2070c0b60fc1c10c6abc7postgres:15-alpine]
docker_network.microshop_net: Refreshing state... [id=cc7a9373c9a0f0077fb33e635d671dbaa0a62aaf7624daf7fb7f2a194fed1760]
docker_container.prometheus: Refreshing state... [id=c5a4126c07e2ca3093263116a604aedd117240c79150e437db2cafb5d97361ca]
docker_container.postgres: Refreshing state... [id=77cb9333a38a11f265369538b5ef766f7f266f0fb26f5e6ab8ebf8b32f5e6507]
docker_container.grafana: Refreshing state... [id=39ee0065e388d3e4adb07ca613de55db7a85f5825f09617d1c7dbe531179ca67]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform planned the following actions, but then encountered a problem:

  # aws_instance.microshop will be created
  + resource "aws_instance" "microshop" {
      + ami                                  = "ami-0c02fb55956c7d316"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + cpu_core_count                       = (known after apply)
      + cpu_threads_per_core                 = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t2.micro"
      + ipv6_address_count                   = (known after apply)
      + ipv6_addresses                       = (known after apply)
      + key_name                             = (known after apply)
      + monitoring                           = (known after apply)
      + outpost_arn                          = (known after apply)
      + password_data                        = (known after apply)
      + placement_group                      = (known after apply)
      + placement_partition_number           = (known after apply)
      + primary_network_interface_id         = (known after apply)
      + private_dns                          = (known after apply)
      + private_ip                           = (known after apply)
      + public_dns                           = (known after apply)
      + public_ip                            = (known after apply)
      + secondary_private_ips                = (known after apply)
      + security_groups                      = (known after apply)
```

Figure: terraform plan - aws_security_group and aws_instance to be created

PROBLEMS DEBUG CONSOLE OUTPUT TERMINAL PORTS

```
      + security_groups                      = (known after apply)
      + source_dest_check                    = true
      + spot_instance_request_id             = (known after apply)
      + subnet_id                            = (known after apply)
      + tags                                 = {
          + "Name" = "microshop-sre-instance"
        }
      + tags_all                             = {
          + "Name" = "microshop-sre-instance"
        }
      + tenancy                              = (known after apply)
      + user_data                            = "2faa7d128b9ce78d681934d1702d56c609b74c55"
      + user_data_base64                     = (known after apply)
      + user_data_replace_on_change          = false
      + vpc_security_group_ids               = (known after apply)

      + capacity_reservation_specification (known after apply)

      + cpu_options (known after apply)

      + ebs_block_device (known after apply)

      + enclave_options (known after apply)

      + ephemeral_block_device (known after apply)

      + instance_market_options (known after apply)

      + maintenance_options (known after apply)

      + metadata_options (known after apply)

      + network_interface (known after apply)

      + private_dns_name_options (known after apply)

      + root_block_device (known after apply)
    }

  # aws_security_group.microshop_sg will be created
  + resource "aws_security_group" "microshop_sg" {
      + arn                    = (known after apply)
      + description            = "Security group for MicroShop SRE assignment"
      + egress                 = [
          + {
              + cidr_blocks      = [
                  + "0.0.0.0/0",
                ]
              + from_port        = 0
              + ipv6_cidr_blocks = []
              + prefix_list_ids  = []
```

Figure: terraform plan - EC2 instance configuration detail

```
Nurzhan@Nurzhan MINGW64 ~/Favorites/Basa/Education/Programming/Introduction to SRE/assignment-4-5/terraform
$ terraform apply -auto-approve
aws_security_group.microshop_sg: Refreshing state... [id=sg-0091acf407323802f]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_instance.microshop will be created
  + resource "aws_instance" "microshop" {
      + ami                                  = "ami-0c02fb55956c7d316"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + cpu_core_count                       = (known after apply)
      + cpu_threads_per_core                 = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t3.micro"
      + ipv6_address_count                   = (known after apply)
      + ipv6_addresses                       = (known after apply)
      + key_name                             = (known after apply)
      + monitoring                           = (known after apply)
      + outpost_arn                          = (known after apply)
      + password_data                        = (known after apply)
      + placement_group                      = (known after apply)
      + placement_partition_number           = (known after apply)
      + primary_network_interface_id         = (known after apply)
      + private_dns                          = (known after apply)
      + private_ip                           = (known after apply)
      + public_dns                           = (known after apply)
      + public_ip                            = (known after apply)
      + secondary_private_ips                = (known after apply)
      + security_groups                      = (known after apply)
      + source_dest_check                    = true
      + spot_instance_request_id             = (known after apply)
      + subnet_id                            = (known after apply)
      + tags                                 = {
          + "Name" = "microshop-sre-instance"
```

Figure: terraform apply - EC2 instance creation in progress

```
Changes to Outputs:
  ~ grafana_url         = "http://localhost:3001" -> (known after apply)
  + instance_public_dns = (known after apply)
  + instance_public_ip  = (known after apply)
  ~ prometheus_url      = "http://localhost:9091" -> (known after apply)
aws_instance.microshop: Creating...
aws_instance.microshop: Still creating... [00m10s elapsed]
aws_instance.microshop: Creation complete after 15s [id=i-0123ea19e003650eb]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:

grafana_url = "http://34.238.137.187:3000"
instance_public_dns = "ec2-34-238-137-187.compute-1.amazonaws.com"
instance_public_ip = "34.238.137.187"
prometheus_url = "http://34.238.137.187:9090"
security_group_id = "sg-0091acf407323802f"

Nurzhan@Nurzhan MINGW64 ~/Favorites/Basa/Education/Programming/Introduction to SRE/assignment-4-5/terraform
$ 
```

Figure: terraform apply - security group created

```
PROBLEMS    DEBUG CONSOLE    OUTPUT    TERMINAL    PORTS

      + spot_instance_request_id             = (known after apply)
      + subnet_id                            = (known after apply)
      + tags                                 = {
          + "Name" = "microshop-sre-instance"
        }
      + tags_all                             = {
          + "Name" = "microshop-sre-instance"
        }
      + tenancy                              = (known after apply)
      + user_data                            = "2faa7d128b9ce78d681934d1702d56c609b74c55"
      + user_data_base64                     = (known after apply)
      + user_data_replace_on_change          = false
      + vpc_security_group_ids               = [
          + "sg-0091acf407323802f",
        ]

      + capacity_reservation_specification (known after apply)

      + cpu_options (known after apply)

      + ebs_block_device (known after apply)

      + enclave_options (known after apply)

      + ephemeral_block_device (known after apply)

      + instance_market_options (known after apply)

      + maintenance_options (known after apply)

      + metadata_options (known after apply)

      + network_interface (known after apply)

      + private_dns_name_options (known after apply)

      + root_block_device (known after apply)
    }

Plan: 1 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  ~ grafana_url         = "http://localhost:3001" -> (known after apply)
  + instance_public_dns = (known after apply)
  + instance_public_ip  = (known after apply)
  ~ prometheus_url      = "http://localhost:9091" -> (known after apply)
aws_instance.microshop: Creating...
aws_instance.microshop: Still creating... [00m10s elapsed]
aws_instance.microshop: Creation complete after 15s [id=i-0123ea19e003650eb]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
```

Figure: terraform apply - outputs: public IP 34.238.137.187

Why Terraform over Manual AWS Console

- Reproducible - identical infrastructure on every terraform apply
- Version controlled - infrastructure changes tracked in git
- Self-documenting - .tf files describe exactly what exists and why
- Automated - no manual clicking, no missed steps
- Destroyable - terraform destroy removes everything, preventing surprise AWS charges

Deployment Guide

Prerequisites

- Docker Desktop installed and running
- Terraform >= 1.0 installed
- AWS account with IAM user (AdministratorAccess, access key + secret key)
- Git

1. Clone and Configure

```
git clone https://github.com/Nurlybek-Nurzhan/Assignment_4-5_SRE.git
cd Assignment_4-5_SRE
cp .env.example .env  # then edit .env with your secrets
```

2. Run Locally

```
docker compose up --build
```

Dashboard: http://localhost:80 | Prometheus: http://localhost:9090 | Grafana: http://localhost:3000

Login with credentials from .env (default: nurzhan / password123).

3. Provision AWS Infrastructure

```
cd terraform
```

Set allowed_ssh_cidr in terraform.tfvars to your IP (e.g. 203.0.113.5/32)

Configure credentials: edit ~/.aws/credentials with your IAM access key

```
terraform init && terraform plan && terraform apply
```

4. Simulate the Incident

In docker-compose.yml change DB_HOST: postgres → DB_HOST: wrong-host

```
docker compose up -d --force-recreate order-service
```

Watch dashboard, check /health, observe order_errors_total in Prometheus.

5. Recover

Restore DB_HOST: postgres in docker-compose.yml

```
docker compose up -d --force-recreate order-service
```

6. Clean Up AWS

```
cd terraform && terraform destroy
```